INCH POUND

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SUPERSEDING
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DETAIL SPECIFICATION

WIRE, STEEL, HIGH CARBON, SQUARE, UNCOATED, FOR MECHANICAL SPRINGS (GENERAL PURPOSE)

Inactive for new design after 16 June 1997

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of uncoated, square (0.050 to 0.500 inches in width), high-carbon steel wire (see 6.1).

1.2 Classification. Wire should be of the following types as specified (see 6.2):

Type I — Oil tempered spring wire for the manufacture of springs which require no subsequent hardening.

Type II — Hand drawn wire for the manufacture of springs which require no subsequent hardening

Type III[1] — Soft (0.55 to 0.75 carbon) wire for the manufacture of springs which are to be hardened after cold forming, furnished in two conditions as follows (see 6.1.3).

Condition A – Annealed at finished size

Condition B – Annealed and lightly drawn.

[1]If a condition letter is not specified for type III wire, either condition A or condition B may be selected at the option of the spring manufacturer.

2. APPLICABLE PUBLICATIONS

2. 1 General. The publications listed in this section are specified in section 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to DLA Troop Support – Construction and Equipment (C&E) (ATTN: Code FCSB), 700 Robbins Avenue Philadelphia, PA 19111-5096 or e-mail to CE_Standardization@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 9505

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-A370 - Standard Test Methods and Definitions for Mechanical Testing of Steel Products
ASTM-A751 - Standard Test Methods and Practices for Chemical Analysis of Steel Products

(Copies of these publications are available from https://www.astm.org.)

SAE INTERNATIONAL (SAE)

SAE-AMS2259 - Chemical Check Analysis Limits Wrought Low-Alloy and Carbon Steels

(Copies of this publication are available from https://www.sae.org.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Process. The steel for the manufacture of the wire shall be made by the open-hearth, basic oxygen, or electric-furnace method.

3.2 Chemical composition. The composition of the steel shall be as shown in Table I, subject to the variations for check analysis given in table 1 of SAE-AMS2259.

TABLE I. Chemical composition. (Ladle analysis)

Type	Carbon Percent	Manganese Percent	Silicon Percent	Phosphorus (max.) Percent	Sulfur (max.) Percent
I and III	0.55 - 0.75	0.60 - 1.20[1]	0.10 - 0.30	0.040	0.050
II	0.45 - 0.75[2]	0.60 - 1.20[3]	0.10 - 0.30	0.040	0.050

[1]For type III wire, manganese shall be between 0.80 and 1.20 percent for widths 0.188 inch and larger.
[2]Not varying more than 0.20 percent in any one lot.
[3]Not varying more than 0.30 percent in any one lot.

3.3 Defects. The wire shall be free from cracks, pipes, and other detrimental imperfections.

3.4 Surface. All wire shall be smooth and free from rust. Type I and type III, condition A, wire shall be free from excessive surface oxidation and shall be oiled or given an equivalent protective coating. Type II and type III, condition B, wire shall be free from scale or surface oxidation and shall be oiled.

3.5 Uniformity. All wire shall be sufficiently uniform in quality and temper to permit manufacture of springs. The wire shall not be wavy or kinked.

3.6 Dimensions. The wire shall be of the size specified, subject to tolerances given in tables II and III.

3.7 Wrapping properties. The wire shall be free from cracks after wrapping around a mandrel of the size shown in table IV. Unless otherwise specified (see 6.4), the wrapping test shall not be required for wire of widths over 0.312 inch or for type III material.

3.8 Mechanical properties.

3.8.1 Type I wire. Type I wire shall be quenched and tempered to obtain the tensile strength specified in table V.

3.8.2 Type II wire. Type II wire shall be cold drawn to obtain the tensile strength specified in table VI.

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TABLE II. Tolerances for width and out-of-square. (All dimensions are in inches)

Width	Tolerance plus and minus	Out-of-square maximum
0.050 to 0.071, incl.	0.0015	0.0020
0.072 to 0.375, incl.	0.0030	0.0040
0.376 to 0.500, incl.	0.0045	0.0060

TABLE III. Corner radii. (All dimensions are in inches)

Width	Corner radii	
	Minimum	Maximum
0.050 to 0.125, incl.	0.012	0.017
0.126 to 0.188, incl.	0.020	0.025
0.189 to 0.250, incl.	0.025	0.035
0.251 to 0.312, incl.	0.030	0.040
0.313 to 0.500, incl.	0.035	0.045

TABLE IV. Mandre size for wrapping test. (All dimensions are in inches)

Wire width	Diameter of mandrel
0.050 to 0.162, incl.	Diagonal of wire
0.163 to 0.312, incl.	Twice the diagonal of wire

TABLE V. Tensile strength, Type I Wire.

Width[1] (in.)	Tensile strength (psi)	
	Minimum	Maximum
0.050	253,000	283,000
0.054	247,000	277,000
0.062	241,000	271,000
0.072	235,000	265,000
0.080	230,000	260,000
0.091	225,000	255,000
0.105	220,000	250,000
0.120	215,000	240,000
0.135	210,000	235,000
0.148	205,000	230,000
0.162	200,000	225,000
0.177	195,000	220,000
0.192	190,000	215,000
0.207	188,000	213,000
0.225	185,000	210,000
0.250	183,000	208,000
0.312	180,000	205,000
0.375	175,000	200,000
0.437	170,000	195,000
0.500	165,000	190,000

[1]For widths other than those shown in table V, tensile strength shall be determined by interpolation.

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TABLE VI. Tensile strength, Type II Wire.

Width[1] (in.)	Tensile strength (psi)	
	Minimum	Maximum
0.050	243,000	279,000
0.054	237,000	272,000
0.062	232,000	266,000
0.072	227,000	261,000
0.080	221,000	254,000
0.091	216,000	248,000
0.105	210,000	241,000
0.120	206,000	237,000
0.135	203,000	234,000
0.148	200,000	230,000
0.162	195,000	225,000
0.177	192,000	221,000
0.192	190,000	218,000
0.207	186,000	214,000
0.225	182,000	210,000
0.250	174,000	200,000
0.312	167,000	193,000
0.375	162,000	187,000
0.437	156,000	180,000
0.500	152,000	176,000

[1]For widths other than those shown in table VI, tensile strength shall be determined by interpolation.

3.8.3 Type III wire.

3.8.3.1 Condition A. Type III, condition A, wire shall be annealed at finished size and shall have no cold work in the finished condition. The maximum tensile strength shall be 95,000 psi.

3.8.3.2 Condition B. Type III, condition B, wire shall be annealed and shall be given at least one pass through dies or rolls after the last annealing treatment. The minimum tensile strength shall be 100,000 psi and the maximum, 130,000 psi.

3.8.3.3 When heat treated. Type III wire, when heat treated, shall develop the properties of table V.

3.9 Cast. Type II wire shall be properly cast. When tested as specified in 4.4.4, wire shall lie substantially flat on itself and shall not spring up and show a wavy condition.

3.10 Coil sizes and weights. Unless otherwise specified, wire shall be furnished in the coil diameters and weights specified in table VII for the appropriate wire size.

3.11 Identification marking. Each coil shall have two substantial tags, printed with the name or trademark of the manufacturer, the specification number, the size, the type, and the condition.

3.12 Workmanship. The wire shall be substantially uniform in quality and temper and shall not have imperfections of a nature or degree, for the grade and quality ordered, that will be detrimental to the forming or fabrication of finished parts.

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TABLE VII. Approximate coil diameters and weights.

Wire type	Wire size (in.)	Approx. mean Coil diameter (in.)	Approx. coil Weights (lb.)
I	0.050 - 0.080	24, 22, 16	75 - 300
	0.081 - 0.160	54, 48, 42, 36, 30	125 - 600
	0.161 - 0.375	60, 54, 48, 42, 36	150 - 600
	0.376 - 0.500	72, 60, 54	150 - 600
II and III	0.050 - 0.0625	22, 16	75 - 300
	0.063 - 0.1875	30, 28, 26, 24, 22	125 - 600
	0.188 - 0.375	36, 30, 28, 26, 24, 22	150 - 600
	0.376 - 0.500	36, 30, 28, 26	150 - 600

4. VERIFICATION

4.1 Conformance inspection. Inspection of the wire is classified as conformance inspection (see 4.2 through 4.4.4).

4.2 Lot. A lot shall consist of wire of the same size, type, condition, and composition, manufactured from the same heat of steel.

4.3 Sampling.

4.3.1 Sampling for chemical tests. When specified, three samples shall be taken from each heat for chemical tests. Specimens for chemical analysis shall be in the form of wire clippings, free from oil, dirt, or other foreign material and shall be representative of the full cross section of the wire. (It is preferable that the product check analysis be performed earlier, such as in the rod or billet stage.)

4.3.2 Sampling for other tests. Five sample coils shall be selected at random from each lot. Specimens shall be taken from one end of each sample coil for tensile, cast (type II and type III, condition B), and wrapping tests.

4.4 Tests.

4.4.1 Chemical tests. Chemical tests shall be made in accordance with ASTM-A751. The samples shall be analyzed for conformance to 3.2.

4.4.2 Tensile test. Tensile strength tests shall be made in accordance with ASTM-A370. The test specimen shall be straight, and the length of the specimen between the jaws of the testing machine shall be not less than 10 inches. Unless otherwise specified, the crosshead speed of the testing machine shall be not less than 0.25 inch per minute and not more than 5 inches per minute.

4.4.3 Wrapping test. Wire small enough to be wrapped by hand may be tested by bending one end of the wire over itself to form a loop with ends crossing at right angles, the shorter end being several inches long, and then tightly winding the longer end five complete turns about the other with turns close together. For larger wire or optionally for wire of any size, the wrapping may be done by any hand or power device which will wrap the wire closely about a mandrel of the required diameter (see 3.7) for five complete turns. The sample shall be considered to have failed if any cracks occur in the wire after the first complete turn. A test shall be repeated if a crack occurs in the first turn (since the wire may be bent locally to a radius less than that specified).

4.4.4 Cast test (type II and type III, condition B). A sample consisting of approximately two full turns of wire carefully cut and removed from a coil shall be placed on a flat surface and examined for conformance to 3.9.

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5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from then managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. This specification covers general purpose carbon steel spring wire intended for springs not subjected to severe repeated stress. Other quality wire, such as music spring wire or valve spring quality wire, should be used where freedom from surface defects and decarburization are required to insure maximum fatigue life for springs operating under repetitive stress.

6.1.1 Type I. The mechanical properties of type I wire are obtained by heat treatment. Springs formed from type I wire are used without further heat treatment, except for suitable low-temperature treatment to relieve residual stresses caused by coiling. Springs formed from type I wire can generally be held to closer dimensional tolerances than springs formed from type II wire.

6.1.2 Type II. The mechanical properties of type II wire are obtained by cold working. Springs formed from type II wire are used without further heat treatment, except for suitable low-temperature treatment to relieve residual stresses caused by coiling.

6.1.3 Type III. Type III wire is a soft, spheroidized annealed wire used where springs must be formed of soft wire. Type III, condition A, wire is generally suitable for more severe deformation than type III, condition B, wire. Springs should be heat treated to specified mechanical properties after forming. The composition of type III is the same as type I.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of the specification.
b. Type of wire.
c. For type III wire, the condition (see 1.2).
d. Width of wire.
e. Quantity of wire.
f. Coil size and weight of wire where desired (see 3.10).
g. Packaging requirements (see 5.1).

6.3 Transportation information. Transportation information applicable to this item is as follows:

Wire, steel, plain
Carload minimum weight 40,000 pounds
Motor volume minimum weight 36,000 pounds.

6.4 Wrapping test. Wire larger than 0.312 inches in width should withstand wrapping around a mandrel equivalent to three times the nominal diagonal. This test is generally not specified, however, because of the difficulty in conducting the test with wrapping equipment normally available.

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6.5 Subject term (key word) listing.

Annealed
Cold forming
Hard drawn
Oil tempered
Soft

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - MR
Navy - OS
DLA - CQ

Preparing activity:
DLA - CQ

(Project 9505-2026-002)

Review activities:
Army - AR, GL, MI

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